

2. (a) The escape velocity, v , of a mass, m , from a spherical mass, M , and radius, R , can be calculated using:

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0$$

- (i) Explain how this equation is an application of conservation of energy. [3]

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- (ii) Calculate the escape velocity from the Sun ($M_{\text{Sun}} = 1.99 \times 10^{30} \text{ kg}$, $R_{\text{Sun}} = 6.96 \times 10^8 \text{ m}$). [3]

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- (b) (i) The temperature of the surface of the Sun is 5780 K. Use a kinetic theory equation to show that the rms speed of a free electron on the surface of the Sun is approximately 500 km s^{-1} . [4]

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- (ii) By considering your answers to (a)(ii) and (b)(i), explain why the Sun has a slight positive charge. [2]

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- (iii) A student claims that a positive charge of approximately 0.08 C on the Sun is enough to produce an electrostatic force equal to the gravitational force on an escaping electron. Determine whether or not she is correct. [3]

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- (iv) Estimate the percentage of lost electrons compared with the total number of electrons on the Sun. Assume that the Sun is mainly hydrogen and that it has lost 0.08 C of charge in the form of electrons. [3]

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